# LnStructured#

https://pytorch.org/docs/stable/generated/torch.nn.utils.prune.LnStructured.

**Description:**

Prune entire (currently unpruned) channels in a tensor based on their Ln-norm. amount (int or float) – quantity of channels to prune. If float, should be between 0.0 and 1.0 and represent the fraction of parameters to prune. If int, it represents the absolute number of parameters to prune. n (int, float, inf, -inf, 'fro', 'nuc') – See documentation of valid entries for argument p in torch.norm(). dim (int, optional) – index of the dim along which we define channels to prune. Default: -1. Add pruning on the fly and reparametrization of a tensor.

## Function Signature

```
class torch.nn.utils.prune.LnStructured(amount, n, dim=-1)
[source]#

Parameters

classmethod apply(module, name, amount, n, dim,
importance_scores=None)[source]#

Parameters

apply_mask(module)[source]#
```

**Returns:**

pruned version of tensor t.

## Notes & Warnings

NOTE Note Pruning itself is NOT undone or reversed!

## Detailed Documentation

### Documentation

Prune entire (currently unpruned) channels in a tensor based on their Ln-norm.

amount (int or float) – quantity of channels to prune. If float, should be between 0.0 and 1.0 and represent the fraction of parameters to prune. If int, it represents the absolute number of parameters to prune.

n (int, float, inf, -inf, 'fro', 'nuc') – See documentation of valid entries for argument p in torch.norm().

dim (int, optional) – index of the dim along which we define channels to prune. Default: -1.

Add pruning on the fly and reparametrization of a tensor.

Adds the forward pre-hook that enables pruning on the fly and the reparametrization of a tensor in terms of the original tensor and the pruning mask.

module (nn.Module) – module containing the tensor to prune

name (str) – parameter name within module on which pruning will act.

amount (int or float) – quantity of parameters to prune. If float, should be between 0.0 and 1.0 and represent the fraction of parameters to prune. If int, it represents the absolute number of parameters to prune.

n (int, float, inf, -inf, 'fro', 'nuc') – See documentation of valid entries for argument p in torch.norm().

dim (int) – index of the dim along which we define channels to prune.

importance_scores (torch.Tensor) – tensor of importance scores (of same shape as module parameter) used to compute mask for pruning. The values in this tensor indicate the importance of the corresponding elements in the parameter being pruned.

If unspecified or None, the module parameter will be used in its place.

Simply handles the multiplication between the parameter being pruned and the generated mask.

Fetches the mask and the original tensor from the module and returns the pruned version of the tensor.

module (nn.Module) – module containing the tensor to prune

pruned version of the input tensor

pruned_tensor (torch.Tensor)

Compute and returns a mask for the input tensor t.

Starting from a base default_mask (which should be a mask of ones if the tensor has not been pruned yet), generate a mask to apply on top of the default_mask by zeroing out the channels along the specified dim with the lowest Ln-norm.

t (torch.Tensor) – tensor representing the parameter to prune

default_mask (torch.Tensor) – Base mask from previous pruning iterations, that need to be respected after the new mask is applied. Same dims as t.

mask to apply to t, of same dims as t

IndexError – if self.dim >= len(t.shape)

Compute and returns a pruned version of input tensor t.

According to the pruning rule specified in compute_mask().

t (torch.Tensor) – tensor to prune (of same dimensions as default_mask).

importance_scores (torch.Tensor) – tensor of importance scores (of same shape as t) used to compute mask for pruning t. The values in this tensor indicate the importance of the corresponding elements in the t that is being pruned. If unspecified or None, the tensor t will be used in its place.

default_mask (torch.Tensor, optional) – mask from previous pruning iteration, if any. To be considered when determining what portion of the tensor that pruning should act on. If None, default to a mask of ones.

pruned version of tensor t.

Remove the pruning reparameterization from a module.

The pruned parameter named name remains permanently pruned, and the parameter named name+'_orig' is removed from the parameter list. Similarly, the buffer named name+'_mask' is removed from the buffers.

Pruning itself is NOT undone or reversed!